

CALIFORNIA STATE UNIVERSITY, NORTHRIDGE

From Social Proof to Action: Algorithmic Amplification as a Conditional Mechanism with
Reach Saturation and Authority Effects

A Graduate Project submitted in partial fulfilment of the requirements

For the degree of Master of Science in Business Analytics

By

Kiran Kanth Madigani

May 2026

(Official end of the semester)

Signature Page

The graduate project of Kiran Kanth Madigani is approved:

Professor Akash Gupta.

Date

Professor Sheng, Qiuhua

Date

Professor Pouyan, Seyed Eslami, Chair

Date

California State University, Northridge

Table of Contents

Signature Page	ii
TOC	iii
Abstract	iv

ABSTRACT

From Social Proof to Action: Algorithmic Amplification as a Conditional Mechanism with Reach Saturation and Authority Effects

By Kiran Kanth Madigani

Master of Science in Business Analytics

Social media sites do more than just show us things; they help decide what gets seen by a lot of people. They use formulas to figure out which posts get a lot of attention. While prior work shows that early engagement can signal value, less is known about how those early persuasive signals are translated into algorithmic amplification, whether amplification produces high-effort engagement rather than superficial attention, and whether additional reach begins to lose effectiveness. This project addresses three questions: how algorithmic amplification translates early persuasive signals into meaningful, high-effort engagement outcomes and whether influencer authority shapes that process; at what point the marginal effect of reach on high-effort engagement diminishes; and whether very high visibility reduces the importance of persuasive power in generating high-effort engagement.

Using a dataset of 10,000 social media posts from 800 influencers, the project operationalizes early persuasive signals through first-hour engagement measures and a persuasive power index, algorithmic amplification through reach and impression metrics, and meaningful outcomes through link clicks, saves, profile visits, composite engagement-success index. The analysis uses regression-based mediation and moderation models with log-transformed variables and clustered standard errors by influencer, along with nonlinear reach specifications to identify saturation effects.

The study provides an improved rationale for how social evidence on algorithms is translated into outcomes of engagement by demonstrating the difference between a widely accepted signal that motivates strong action. The expected findings are that higher levels of early persuasive signals will be correlated with higher levels of amplification, that amplification will be correlated with higher levels of high-effort engagement, and that authority from influencers will enhance the relationship. Also, it is expected that as influence reaches a point where it has been amplified to a great extent, there will be diminishing returns, so that at some level of high visibility, the degree of predictability of early persuasive signal strength will lead to smaller increases in meaningful engagement.

Keywords: algorithmic amplification; influencer authority; reach saturation; high-effort engagement